Project - Implementation (Part 3)

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Case Study: Reliable Rentals

A company called Reliable Rentals rents out vehicles at different outlets (locations/sites). Each outlet has a number, address, phone number and a fax number. Each site is allocated a stock of vehicles for hire. The registration number uniquely identifies each vehicle for hire and is used when hiring a vehicle to a client.

Clients may hire vehicles for various periods of time. Each individual hire agreement between a client and the Company is uniquely identified using a hire number. Information stored on the vehicles for hire include: the vehicle registration number, model, make, engine size, capacity, current mileage, daily hire rate, and the current location (outlet) of each vehicle. The data stored on clients includes the client number (unique identifier), name (first and last name), home address, phone number, date of birth, and driving license number.

The data stored on a hire agreement includes the hire number, the client's number, name, address, and phone number, date the client started the hire period, date the client wishes to terminate the hire period, the vehicle registration number, model and make, the mileage before and after the hire period. Finally, information is stored on the staff based at various outlets including: staff number, name (first and last name), home address, home phone number, date of birth (DOB), sex, date joined the Company, job title, and salary. Each staff member is associated with a single outlet.

Implement the database using embedded SQL (Python and SQLite as in the assignment).

A. Develop SQL code to create the entire database schema, reflecting the constraints identified in previous steps.

<u>Constraints:</u>

```
--- Table: Client ---
CREATE TABLE Client(
    clientNo VARCHAR(100) NOT NULL,
    fName VARCHAR(100) NOT NULL,
    lName VARCHAR(100) NOT NULL,
```

```
    home_address VARCHAR(200) NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    DOB DATE NOT NULL,
    driving_license_number VARCHAR(50) NOT NULL UNIQUE,

    PRIMARY KEY(clientNo)
    )

--- Table: HireAgreement ---
CREATE TABLE HireAgreement(
    hireNo VARCHAR(100) NOT NULL,
    clientNo VARCHAR(100) NOT NULL,
    registrationNo VARCHAR(100) NOT NULL,
    date_of_hire DATE NOT NULL,
    hire_termination_date DATE NOT NULL,
    mileage_before INT NOT NULL CHECK(mileage_before >= 0),
    mileage_after INT,

    PRIMARY KEY(hireNo),
    FOREIGN KEY(clientNo) REFERENCES Client(clientNo),
    FOREIGN KEY(registrationNo) REFERENCES Vehicle(registrationNo),
    CHECK(date_of_hire <= hire_termination_date),
    CHECK(mileage_after IS NULL OR mileage_after >= mileage_before)
    )

--- Table: Outlet ---
CREATE TABLE Outlet(
    outletNo VARCHAR(100) NOT NULL,
    address VARCHAR(200) NOT NULL,
    phone_number VARCHAR(20) NOT NULL UNIQUE,
    fax_number VARCHAR(20) NOT NULL UNIQUE,
```

```
    PRIMARY KEY(outletNo)
    )

--- Table: Staff ---
CREATE TABLE Staff(
    staffNo VARCHAR(100) NOT NULL,
    outletNo VARCHAR(100) NOT NULL,
    fName VARCHAR(100) NOT NULL,
    lName VARCHAR(100) NOT NULL,
    home_address VARCHAR(200) NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    DOB DATE NOT NULL,
    sex VARCHAR(1) NOT NULL CHECK(sex IN ('M', 'F')),
    company_start_date DATE NOT NULL,
    job_title VARCHAR(100) NOT NULL,
    salary REAL NOT NULL CHECK(salary > 0),

    PRIMARY KEY(staffNo),
    FOREIGN KEY(outletNo) REFERENCES Outlet(outletNo),
    CHECK(DATE(company_start_date) >= DATE(DOB, '+16 years'))
    )

--- Table: Vehicle ---
CREATE TABLE Vehicle(
    registrationNo VARCHAR(100) NOT NULL,
    outletNo VARCHAR(100) NOT NULL,
    model VARCHAR(100) NOT NULL,
    make VARCHAR(100) NOT NULL,
    engine_size REAL NOT NULL CHECK(engine_size > 0),
    capacity INT NOT NULL CHECK(capacity >= 1),
```

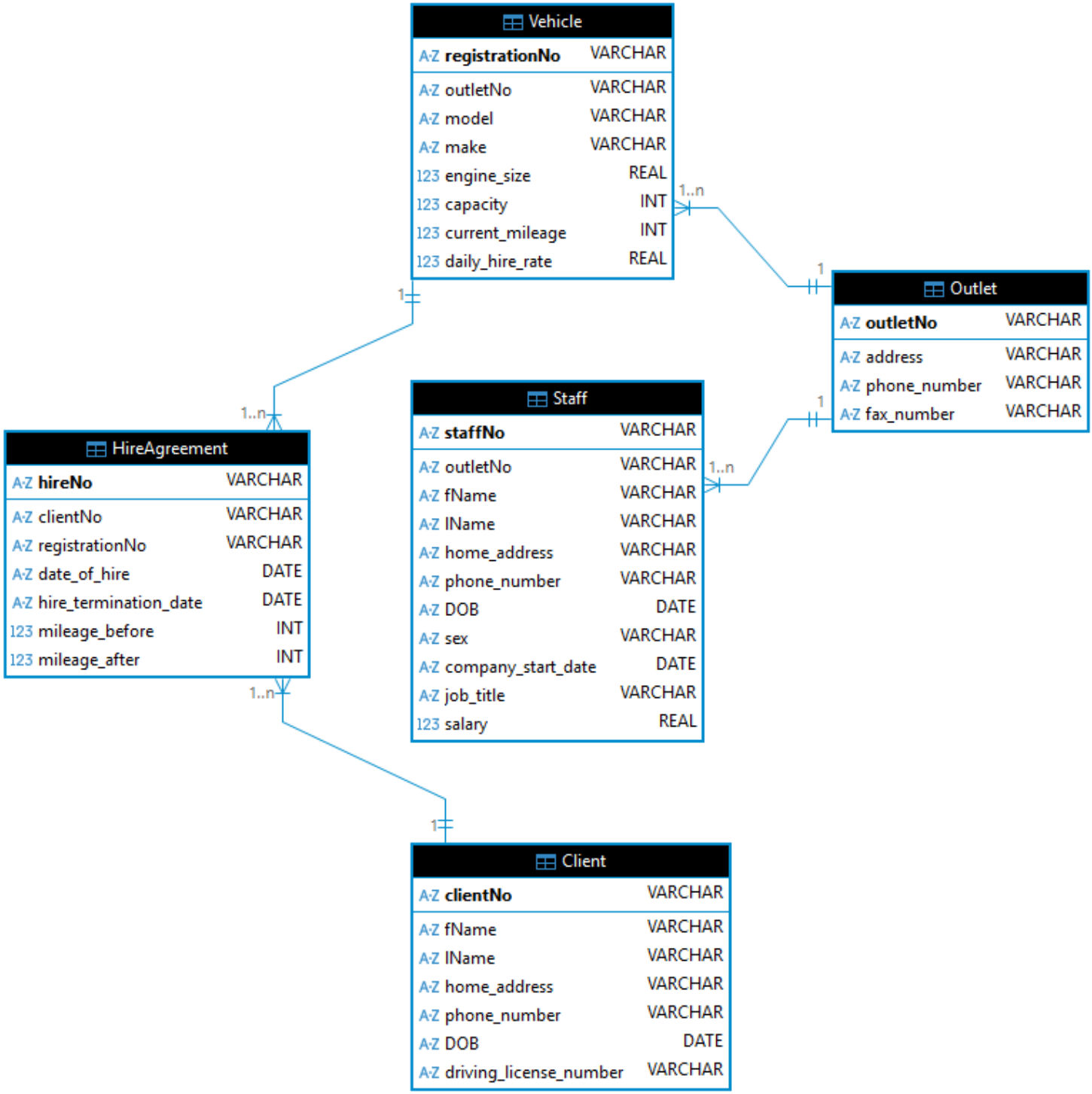
```
current_mileage INT NOT NULL CHECK(current_mileage >= 0),
daily_hire_rate REAL NOT NULL CHECK(daily_hire_rate > 0),

PRIMARY KEY(registrationNo),
FOREIGN KEY(outletNo) REFERENCES Outlet(outletNo)
)
```

Unable to implement the following constraints:

Constraint	**Relation(s)**	**Reason**
DOB must be a past date	Client, Staff	SQLite does not allow DATE('now') in CHECK (non-deterministic)
Client must be ≥ 18 at hire signing	Client, HireAgreement	Cross-table + time-dependent; cannot be expressed in CHECK
Only one active hire per vehicle	HireAgreement	CHECK cannot reference other rows in the same table
Each outlet has ≥ 1 staff	Outlet, Staff	Cross-table; CHECK cannot reference other tables. Satisfied through sample data
mileage_before matches vehicle's current_mileage at hire time	HireAgreement, Vehicle	Cross-table; CHECK cannot reference other tables

Diagram:

Note: *Embedded SQL was used to create the database (Python and SQLite), DBeaver was used for visualization purposes only.*



B. Create at least 5 tuples for each relation in your database.

Outlet

outletNo	address	phone_number	fax_number
O001	120 Brickell Ave, Miami, FL	305-555-0101	305-555-0201
O002	450 Collins St, Miami Beach, FL	305-555-0102	305-555-0202
O003	88 SW 8th St, Coral Gables, FL	305-555-0103	305-555-0203
O004	301 Las Olas Blvd, Fort Lauderdale	954-555-0104	954-555-0204
O005	12 Ocean Dr, Cutler Bay, FL	305-555-0105	305-555-0205

Client

clientNo	fName	lName	home_address	phone_number	DOB	driving_license_number
C001	Olivia	Martinez	101 Sunset Dr, Miami	305-555-2001	1992-04-15	D-MART-9201
C002	Ethan	Brown	55 Coral Way, Coral Gables	305-555-2002	1985-08-22	D-BROW-8501
C003	Isabella	Davis	780 Biscayne Blvd, Miami	305-555-2003	1998-12-03	D-DAVI-9801
C004	Noah	Wilson	22 Ocean Ave, Miami Beach	305-555-2004	1979-06-10	D-WILS-7901
C005	Ava	Thompson	300 Las Olas, Fort Lauderdale	954-555-2005	2000-01-28	D-THOM-0001
C006	Lucas	Garcia	14 Marina Rd, Cutler Bay	305-555-2006	1990-10-19	D-GARC-9001

Hire Agreement

hireNo	clientNo	registrationNo	date_of_hire	hire_termination_date	mileage_before	mileage_after
H10001	C002	FL-C9012	2026-01-10	2026-01-17	30,200	30,850
H10002	C004	FL-G6789	2026-02-05	2026-02-12	14,500	14,950
H10003	C001	FL-A1234	2026-04-20	2026-05-05	18,500	[NULL]
H10004	C003	FL-B5678	2026-04-15	2026-05-10	22,000	[NULL]
H10005	C006	FL-F2345	2026-04-10	2026-05-01	27,500	[NULL]
AXNZN	C005	FL-E7890	2026-04-25	2026-05-15	8,200	[NULL]
H10006	C002	FL-H0123	2026-04-27	2026-05-04	9,800	[NULL]

Staff

staffNo	outletNo	fName	lName	home_address	phone_number	DOB	sex	company_start_date	job_title	salary
ST001	O001	Maria	Gonzalez	500 NE 1st Ave, Miami	305-555-1001	1985-03-12	F	2015-06-01	Outlet Manager	62,000
ST002	O002	James	Carter	720 Lincoln Rd, Miami Beach	305-555-1002	1990-07-22	M	2018-09-15	Rental Agent	42,000
ST003	O003	Aisha	Nguyen	15 Granada Blvd, Coral Gables	305-555-1003	1992-11-04	F	2019-01-10	Outlet Manager	58,000
ST004	O004	Daniel	Rivera	88 SE 17th St, Fort Lauderdale	954-555-1004	1988-05-18	M	2016-04-20	Senior Agent	51,000
ST005	O005	Sophia	Kim	210 Marina Dr, Cutler Bay	305-555-1005	1995-09-30	F	2021-02-08	Rental Agent	40,000
ST006	O001	Liam	Patel	350 NW 7th St, Miami	305-555-1006	1993-12-14	M	2020-07-01	Maintenance Tech	38,000

Vehicle

registrationNo	outletNo	model	make	engine_size	capacity	current_mileage	daily_hire_rate
FL-A1234	O001	Camry	Toyota	2.5	5	18,500	55
FL-B5678	O001	Corolla	Toyota	1.8	5	22,000	45
FL-C9012	O002	Civic	Honda	2	5	31,000	50
FL-D3456	O002	C-Class	Mercedes Benz	2	5	12,500	110
FL-E7890	O003	E-Class	Mercedes Benz	3	5	8,200	145
FL-F2345	O003	RAV4	Toyota	2.5	5	27,500	65
FL-G6789	O004	Mustang	Ford	5	4	15,000	120
FL-H0123	O005	Model 3	Tesla	0.1	5	9,800	130

C. Develop the 5 SQL queries that correspond to Part 2.C using embedded SQL.

First User Transaction

What is the registration number, mileage before hire, and client full name for the most recently signed hire agreement?

Query:

```
SELECT ha.registrationNo, ha.mileage_before,
           c.fName || ' ' || c.lName AS client_full_name,
           ha.date_of_hire
    FROM HireAgreement ha, Client c
    WHERE ha.clientNo = c.clientNo
    ORDER BY ha.date_of_hire DESC
    LIMIT 1;
```

Output:

registrationNo	mileage_before	client_full_name	date_of_hire
FL-H0123	9,800	Ethan Brown	2026-04-27

Second User Transaction

What is the highest daily hire rate charged across all hire agreements?

Query:

```
SELECT MAX(v.daily_hire_rate) AS highest_daily_hire_rate
```

```
    FROM Vehicle v, HireAgreement ha

    WHERE v.registrationNo = ha.registrationNo;
```

Output:

123 highest_daily_hire_rate
145

Third User Transaction

Which outlets have vehicles whose current mileage exceeds the average mileage of all vehicles hired in the last 2 months, and what would their updated daily hire rate be after a 5% increase?

Query:

```
SELECT DISTINCT o.outletNo, o.address, v.registrationNo,

           v.current_mileage,

           v.daily_hire_rate AS current_rate,

           ROUND(v.daily_hire_rate * 1.05, 2) AS projected_rate

    FROM Outlet o, Vehicle v, HireAgreement ha

    WHERE v.outletNo = o.outletNo

      AND ha.registrationNo = v.registrationNo

      AND v.current_mileage > (

          SELECT AVG(v2.current_mileage)

          FROM Vehicle v2, HireAgreement ha2

          WHERE ha2.registrationNo = v2.registrationNo

            AND ha2.date_of_hire >= DATE('now', '-2 months')

      )
```

```
    ORDER BY o.outletNo, v.registrationNo;
```

Output:

outletNo	address	registrationNo	current_mileage	current_rate	projected_rate
O001	120 Brickell Ave, Miami, FL	FL-A1234	18,500	55	57.75
O001	120 Brickell Ave, Miami, FL	FL-B5678	22,000	45	47.25
O002	450 Collins St, Miami Beach, FL	FL-C9012	31,000	50	52.5
O003	88 SW 8th St, Coral Gables, FL	FL-F2345	27,500	65	68.25

<u>Fourth User Transaction</u>

Toyota vehicles are being hired very often. Before importing more, how many clients currently have an active hire agreement for a Toyota vehicle, and what are the specific models involved?

Part 1

Query:

```
SELECT COUNT(DISTINCT ha.clientNo) AS active_toyota_clients

    FROM HireAgreement ha, Vehicle v

    WHERE ha.registrationNo = v.registrationNo

      AND v.make = 'Toyota'

      AND DATE('now') BETWEEN ha.date_of_hire AND
ha.hire_termination_date;
```

Output:

active_toyota_clients
3

Part 2

Query:

```
SELECT DISTINCT v.model
```

```
    FROM HireAgreement ha, Vehicle v

    WHERE ha.registrationNo = v.registrationNo

      AND v.make = 'Toyota'

      AND DATE('now') BETWEEN ha.date_of_hire AND ha.hire_termination_date

    ORDER BY v.model;
```

Output:

A-Z model
Camry
Corolla
RAV4

<u>Fifth User Transaction</u>

A client called but the call was dropped. They only provided the hire number, AXNZN, for a Mercedes Benz with the highest engine size. What are their contact details and address?

Query:

```
SELECT c.clientNo,

           c.fName || ' ' || c.lName AS client_full_name,

           c.home_address, c.phone_number,

           v.make, v.model, v.engine_size

    FROM HireAgreement ha, Client c, Vehicle v

    WHERE ha.clientNo = c.clientNo

      AND ha.registrationNo = v.registrationNo

      AND ha.hireNo = 'AXNZN'

      AND v.make = 'Mercedes Benz'

      AND v.engine_size = (SELECT MAX(engine_size) FROM Vehicle WHERE make
= 'Mercedes Benz');
```

Output:

clientNo	client_full_name	home_address	phone_number	make	model	engine_size
C005	Ava Thompson	300 Las Olas, Fort Lauderdale	954-555-2005	Mercedes Benz	E-Class	3

D. Upload all the code and documentation (including previous project submissions) to GitHub.

Notes: A report will be created for each deadline including detailed documentation of each of the steps addressing all the required items. Screenshots of the contents of the database created for each portion of this part must be included in the report and the corresponding queries to show the table contents must be included in the code.

GitHub: The code generated must be uploaded to a GitHub repository. The link to the repository must be provided in the last report. The GitHub repository must also include all the documentation (i.e., reports) generated in the three project parts. The GitHub should be organized containing appropriate folders and a README file with a description of the project should be included.

GitHub Repository:

https://github.com/salome-ramirez/DSC623---Project-DB